

Spatial Determinants of Anime Store Locations in Tokyo: A GIS-Based Framework for Cultural Retail Analysis

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Introduction and Problem Statement

One of the most important decisions any retail business has to make is where to open. Location affects everything from how many customers walk through the door to whether the store is even visible to the right people. In retail geography researchers have long argued that these decisions are not just economic, they are deeply spatial. Businesses respond to the geography around them including things like population density, proximity to competitors, transportation access, and the cultural character of a neighborhood. Geographic Information Systems, or GIS, gives researchers the tools to analyze these patterns in a rigorous and repeatable way.

Previous work has shown how valuable GIS can be for this kind of analysis. Roig-Tierno et al. (2013) point out that despite how useful spatial analysis tools are for identifying retail opportunities, many businesses still do not use them when making site selection decisions. Their study shows that overlaying demographic, competitive, and geographic data in GIS can reveal patterns that conventional market research simply misses. On the clustering side, Wrigley and Lowe (2002) explain that stores rarely spread themselves evenly across a city. Instead they tend to concentrate in areas of high consumer activity, partly because clustering itself attracts more customers. This idea is extended by Scott (2000), who argues that cultural industries in particular form geographic clusters in cities because consumers expect to find related products and experiences in the same place. Over time those expectations become self reinforcing, a district develops a reputation, which draws more stores, which strengthens the reputation further.

Tokyo is a particularly good place to study these dynamics. It has a massive and well documented anime industry that generates billions of dollars in merchandise revenue annually (Yano Research Institute, 2022), and it contains several distinct cultural retail districts that are globally recognized. Akihabara in Chiyoda ward, Ikebukuro in Toshima ward, and Nakano Broadway in Nakano ward are probably the three most well known. These are not just neighborhoods where anime stores happen to exist. They are destinations in their own right, drawing consumers from across Tokyo and from other countries specifically because of their cultural identity. Akihabara in particular transformed over several decades from an electronics market into the global center of anime merchandise, a process Galbraith (2010) traces in detail and that illustrates how cultural retail districts build the reputational capital that makes them self sustaining.

Just looking at a map of anime store locations in Tokyo makes clear that these stores are not spread evenly. They cluster heavily in and around these districts with very few stores in the outer suburban areas. What is less clear is exactly which spatial variables are driving that pattern. Is it

population density? Commercial rent levels? Proximity to established cultural destinations? Or some combination of all three? The answer matters not just for understanding anime retail specifically but for understanding how cultural retail districts form and sustain themselves more generally.

This study uses GIS spatial analysis to examine those questions. Specifically it looks at how population density, commercial rent zones, and proximity to cultural districts relate to the locations of anime retail stores across the Tokyo metropolitan area. A secondary goal is to build a reusable ArcPy automation framework that organizes the entire analytical workflow so that it can be adapted for future studies of cultural retail in other cities. This work is relevant to GIS researchers, urban planners, and anyone interested in how cultural industries organize themselves spatially in major cities.

While GIS has been applied extensively to traditional retail sectors like supermarkets and shopping centers (Hernandez and Bennison, 2000; Baviera-Puig et al., 2016), specialized cultural retail industries have received comparatively little spatial attention. Most existing studies of anime and otaku culture take an ethnographic or cultural studies approach, which is valuable but does not provide the kind of systematic spatial evidence this project generates. This project helps fill that gap by testing whether established spatial frameworks can explain the distribution of a niche but economically significant retail sector, and by building a reusable analytical tool for future researchers.

This project is focused on two main research questions. First, what spatial characteristics explain where anime stores locate in Tokyo, particularly in relation to population density, commercial rent zones, and cultural districts? Second, how can those spatial relationships be organized into a reproducible GIS workflow using ArcPy that could support future cultural retail analysis?

Hypothesis

The hypothesis for this study is that anime retail stores in Tokyo are more likely to be found in areas with higher population density, moderate to high commercial rent levels, and closer proximity to established cultural districts than would be expected if store locations were random. Moderate rent is defined as the middle two tiers in the dataset, roughly 45,000 to 90,000 yen per month, which covers the majority of non-central wards. Cultural districts refer specifically to Akihabara, Ikebukuro, and Nakano Broadway, which are recognized both in the academic literature and among consumers as the primary anime retail destinations in the city.

This hypothesis is testable and falsifiable. By generating a set of random points across the same study area and comparing their spatial characteristics to the real anime store locations, it is possible to determine whether the observed pattern is statistically different from chance. If the real stores are not significantly closer to population centers or cultural districts than the random points, the hypothesis would not hold. The rationale behind the prediction is that anime retail is a specialized cultural industry and should therefore exhibit the clustering behavior that cultural

economic geography theory predicts, with stores gravitating toward districts where cultural reputation and consumer demand already reinforce each other rather than spreading evenly to minimize rent exposure.

Related Literature

The literature relevant to this study spans retail geography, cultural economic geography, GIS based spatial analysis, and Japanese urban studies.

In retail geography, Christaller (1933) established through central place theory that retail businesses tend to locate in ways that maximize market area and minimize overlap with competitors. This classical framework predicts a relatively even distribution of retail across space, calibrated to population. Wrigley and Lowe (2002) update this framework by emphasizing agglomeration economies, where businesses actually benefit from clustering together because shared proximity draws consumers who are already in a shopping mindset and reduces search costs for buyers looking for related products. This agglomeration dynamic is especially relevant to anime retail where the presence of multiple stores in one area reinforces the identity of that district as a destination worth traveling to. Roig-Tierno et al. (2013) demonstrate that overlaying demographic, competitive, and geographic data in GIS can reveal location patterns that conventional market research misses, and argue that more businesses should use spatial tools in their decision making. Hernandez and Bennison (2000) and Baviera-Puig et al. (2016) provide additional applications of GIS to retail location in traditional sectors that inform the methodological approach taken here.

Cultural economic geography provides a complementary theoretical lens. Scott (2000) argues in *The Cultural Economy of Cities* that cultural industries form geographic clusters in urban environments because of the mutual reinforcement between consumption and place identity. Cultural clusters develop reputations that attract both producers and consumers, creating a self reinforcing dynamic that is distinct from the cost minimization logic of classical retail location theory. Flew (2012) extends this to creative industries globally and argues that the clustering of creative and cultural businesses in specific urban districts is a feature of contemporary urban economies worldwide. Pratt (1997) distinguishes between production clusters, where cultural goods are made, and consumption clusters, where they are sold, noting that anime retail districts like Akihabara are primarily consumption clusters with their own spatial logic that is driven more by reputation and consumer expectation than by production cost considerations. Florida (2002) connects cultural clustering to broader urban economic development and argues that cities with strong creative and cultural clusters attract talent and investment in ways that reinforce their competitive position, giving this kind of analysis relevance beyond just anime retail.

On the Japanese side, Galbraith (2010) provides an essential ethnographic account of how Akihabara transformed from an electronics district into the global center of anime merchandise over several decades, tracing the interplay between consumer culture, retail investment, and place identity that produced the current clustering pattern. Morikawa (2003) examines

Ikebukuro's distinct identity as an anime district with its own demographic character, emphasizing that it developed differently from Akihabara and serves a somewhat different consumer base, which helps explain why two statistically significant clusters rather than one dominate the central Tokyo area. Kanemoto and Kurima (2005) find that specialized retail clusters in Tokyo form around major transit hubs, which is directly relevant to this study because both Akihabara and Ikebukuro are among Tokyo's busiest train stations. Sorensen (2002) situates this in the broader history of Japanese urban commercial development and argues that the concentration of specialized retail around transit infrastructure reflects both market forces and planning decisions that have historically shaped Tokyo's commercial geography.

Methodologically, Getis and Ord (1992) introduce the G_i^* spatial statistic that forms the backbone of the hotspot analysis in this study. Their method evaluates whether the local concentration of a phenomenon is significantly higher or lower than would be expected given the global distribution, and provides confidence levels that allow for statistically defensible inference. Mitchell (2005) provides comprehensive guidance on spatial statistics in GIS including near table analysis and distance based comparison methods, which directly inform the methodological choices made in this project.

Data Sources

This analysis uses four primary datasets. Anime store point locations were collected from publicly available online store directories and geocoded for use in ArcGIS Pro. These points represent the known locations of anime merchandise retailers across the Tokyo metropolitan area and constitute the dependent variable in the analysis. Tokyo ward level population density data was obtained from Japanese government census sources and serves as a proxy for consumer demand across the study area, representing the residential population per square kilometer for each ward.

Commercial rent zone polygons were derived from ward level average residential rent data for single room units compiled from Japanese real estate market surveys. True ward level commercial rent data was not readily available so residential rent was used as a spatial proxy, justified by the positive correlation between residential and commercial rents at the neighborhood scale, meaning wards with higher residential rents generally also carry higher commercial costs. This approach is acknowledged as a limitation throughout the analysis and is discussed explicitly in the results section. The rent variable is treated as a proxy for the general economic cost environment rather than a direct measure of commercial lease rates.

Finally a set of randomly generated points was produced within the Tokyo study boundary using the Create Random Points tool in ArcGIS Pro to serve as a null model for comparison against the observed store locations in the near table analysis. The number of random points generated matched the number of real anime store locations to ensure a fair comparison. All datasets were projected into JGD2011 / UTM Zone 54N to preserve distance measurement accuracy across the study area, which is essential for the near table distance calculations and the hotspot analysis.

Methodology

The analysis moves through five connected steps, each building on the last, and the entire workflow was automated using ArcPy scripts to make it reproducible and adaptable for future use.

The first step was mapping the raw distribution of anime stores across Tokyo municipalities (Figure 1) to establish the spatial pattern visually before any statistical testing. This step is important because it motivates the rest of the analysis by confirming that clustering exists and raising the question of which factors explain it. The map shows immediately that stores pile up in a small number of central wards while the outer suburban areas have almost nothing.

The second step was a population density overlay, spatially joining ward level density data to the store point layer and visualizing stores against a density choropleth (Figure 2). The rationale for this step is that retail location theory predicts businesses should locate near population centers to maximize consumer accessibility. By overlaying stores on the density map it is possible to evaluate whether the wards with the most stores are also the most densely populated, or whether other factors are at work.

The third step was a commercial rent zone overlay, spatially joining each store point to its ward level rent polygon and visualizing the distribution across five rent tiers from under 45,000 yen to 90,000 to 131,300 yen per month (Figure 3). The five tier classification was chosen to provide sufficient resolution to distinguish between different economic environments while keeping the map readable. This step tests whether economic cost environment shapes location decisions and whether stores avoid high rent areas as basic cost minimization logic would suggest. Finding the opposite, that stores concentrate in high rent zones, is a substantively interesting result that requires theoretical explanation.

The fourth step was a Near Table distance comparison using the Generate Near Table tool to calculate straight line distances from both real anime stores and randomly generated comparison points to their nearest municipal population centroid, visualized as connecting lines (Figure 4). Blue lines connect real anime stores to their nearest centroid and red lines connect random points to theirs, with yellow points marking the centroids themselves. This is the most directly quantitative step in the analysis. By comparing the distribution of line lengths for real stores versus random points it is possible to determine whether anime stores are systematically closer to population centers than would be expected by chance. Municipal population centroids were used as reference points rather than ward centroids because they more closely approximate the actual geographic center of population mass within each administrative unit.

The fifth step was a Getis-Ord G_i^* hotspot analysis using the Optimized Hot Spot Analysis tool in ArcGIS Pro (Figure 5). G_i^* was chosen over simple density mapping or kernel density estimation because it provides statistical confidence levels rather than just a visual impression of clustering, allowing for defensible inference about which districts represent statistically

significant concentrations. The optimized version of the tool automatically selects the appropriate distance band based on the spatial distribution of the input data, avoiding the need to manually specify a neighborhood distance that could bias the results. The output classifies each area as a hot spot or cold spot at 90%, 95%, or 99% confidence, or as not significant.

The entire workflow was scripted in ArcPy with input paths and key parameters defined as variables at the top of each script. This design makes it straightforward to substitute different input datasets without rewriting the analytical logic. The scripts handle spatial joins using SpatialJoin, distance calculations using GenerateNearTable, random point generation using CreateRandomPoints, hotspot analysis using OptimizedHotSpotAnalysis, and data export for visualization. Each stage is modular meaning individual steps can be updated or swapped out without rebuilding the entire workflow. A researcher studying gaming stores in Osaka or manga cafes in Kyoto would replace the input layers but the analytical logic and tool calls remain the same, which is the core value of the framework as a contribution beyond this particular study.

Results and Discussion

The base distribution map (Figure 1) makes the clustering pattern immediately visible. Stores pile up heavily in Chiyoda, Toshima, and Shibuya while outer suburban wards like Machida have almost nothing, confirming the pattern is non-random before any statistics are run. This visual evidence sets up the central question of which spatial factors explain the concentration.

The population density overlay (Figure 2) shows that the most store-dense wards are urbanized and relatively dense but not always the highest density residential wards in the metro area. Chiyoda and Minato have large daytime commercial populations despite relatively low residential density, suggesting residential population density alone does not fully capture the consumer accessibility that matters for anime retail. Wards like Adachi and Katsushika have relatively high residential density but very few anime stores, which further suggests that raw population concentration is not the primary driver. Commercial foot traffic, transit connectivity, and cultural identity likely matter considerably more than residential headcounts for this type of specialized retail.

The rent zone overlay (Figure 3) shows the most surprising finding in the study. Rather than concentrating in moderate rent areas where costs are more manageable, a large share of stores appear in the highest rent tiers, particularly the 65,000 to 90,000 yen and 90,000 to 131,300 yen ranges centered on central Tokyo wards. The Chiyoda cluster, home to Akihabara, sits in some of the most expensive real estate in the city. This is essentially the opposite of what basic retail location theory would predict. The interpretation is that cultural destination status is a stronger pull factor than cost minimization for this type of retail. Stores are willing to absorb higher rents because being in Akihabara is itself part of the product, it signals credibility to consumers and ensures access to the foot traffic of fans who are already making a trip to the district. This finding aligns with Scott's (2000) argument that the reputational value of being in a recognized cultural district can outweigh the economic cost of co-location. It is worth noting again that the rent data

here reflects residential rather than commercial rents, so the specific figures should be interpreted with that limitation in mind, but the spatial pattern of where stores cluster relative to the rent gradient is still meaningful and interpretable.

The near table analysis (Figure 4) shows that anime stores are consistently positioned closer to municipal population centroids than randomly generated comparison points across both the dense central cluster and the more scattered outer locations. The blue lines connecting real stores to their nearest centroid are visibly shorter than the red lines connecting random points to theirs across nearly every part of the study area. This supports the hypothesis that consumer accessibility is a meaningful spatial predictor of store location, consistent with what retail location theory would predict. Combined with the rent finding, the picture that emerges is that stores are not just clustering wherever population is highest but specifically in districts where cultural identity and consumer demand already overlap and reinforce each other. Accessibility matters but it is not sufficient on its own to explain where stores end up.

The Gi* hotspot results (Figure 5) statistically confirm what the other maps suggest visually. The strongest cluster at 99% confidence is centered on Chiyoda ward, corresponding directly to Akihabara. Two additional clusters appear at 90 to 95% confidence in Toshima ward (Ikebukuro) and Shibuya. No cold spots were detected anywhere in the study area, which is an important finding. It means anime stores are concentrated in specific districts but are not systematically avoiding any particular area. The pattern is one of positive clustering driven by cultural destination identity rather than dual clustering and avoidance. The Ikebukuro and Shibuya results are also worth noting because they represent somewhat different retail environments from Akihabara, suggesting the clustering pattern is not just about one famous district but reflects something broader about how cultural retail organizes itself across the city when multiple districts have developed strong reputational identities.

Taken together these results support the hypothesis. Anime stores cluster in areas with high consumer accessibility and proximity to established cultural districts, and the Gi* analysis gives statistical confirmation of the three main clusters that retail geography and cultural clustering theory both predict. The rent finding adds the most interesting nuance. Stores appear willing to operate in high rent environments when the cultural and commercial advantages of a district are strong enough, which aligns with Scott's (2000) argument that reputational value can outweigh economic cost of co-location. The near table comparison backs this up by showing that proximity to population centers is a real and measurable spatial predictor, but the rent overlay makes clear that cost minimization is not the dominant logic.

Broader Impacts and Significance

This study adds to the literature on retail geography and cultural economic clustering by applying systematic spatial analysis to a sector that has not received much academic attention. The finding that cultural reputation and consumer accessibility outweigh cost minimization as location predictors adds nuance to ongoing theoretical debates about what drives cultural industry

clustering, and the cross-field relevance to urban planning and cultural economics gives the work broader significance.

Outside of academia the research has practical value for urban planners thinking about how to protect or cultivate creative retail corridors, and for small business owners in the cultural retail space trying to make smarter site selection decisions beyond intuition. The ArcPy framework is a practical contribution that could be reused for future studies and has potential as a teaching tool for students learning spatial analysis and reproducible research methods. As more cities compete to cultivate creative economies and cultural tourism, having systematic reusable methods for analyzing how cultural retail organizes itself spatially becomes increasingly useful.

Conclusion and Future Directions

This study confirms that anime store clustering in Tokyo is statistically significant and driven primarily by cultural destination identity and consumer accessibility rather than cost minimization. The Gi* analysis identified three statistically significant clusters in Chiyoda, Toshima, and Shibuya consistent with the known cultural geography of anime in the city. The near table comparison showed real stores locate significantly closer to population centers than random points, and the rent overlay showed stores are concentrated in high rent zones rather than moderate ones.

The main limitation is the use of residential rent as a proxy for commercial rent. Future research should incorporate actual commercial rent data, daytime population estimates, and transit ridership data to better capture within ward variation. Applying the same ArcPy framework to other Japanese cities like Osaka or Kyoto would test whether these patterns generalize beyond Tokyo, which is the most straightforward next step this project enables.

References

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This map of Tokyo illustrates the correlation between average rent prices and the density of anime stores. The city is divided into wards, each shaded in a color representing its average rent price. A legend in the bottom right corner defines the rent price ranges: No Data (white), 0 - 45,000 (lightest green), 45,000 - 65,000 (light green), 65,000 - 90,000 (medium green), and 90,000 - 131,300 (darkest green). Yellow dots represent the locations of anime stores. The map shows a high concentration of stores in the central wards, particularly in the darkest green areas like Chiyoda, Minato, and Nishi-Shinjuku, which also have the highest average rent prices. Other wards like Shibuya and Saitama also show a high density of stores. The map includes labels for various wards and neighboring cities like Maebashi, Utsunomiya, and Yokohama.

Source: Bk1, Taitoh, Ginko, 01 Concept/Strategy, contributors, and the GIS User Community; Soshiki Bk1, Taitoh, Ginko, Bk1, MCAA, JGSE, 01 Concept/Strategy, contributors, and the GIS User Community.

Figure 4. Near table distance comparison: anime stores (blue) vs. randomly generated points (red) relative to municipal population centroids (yellow).

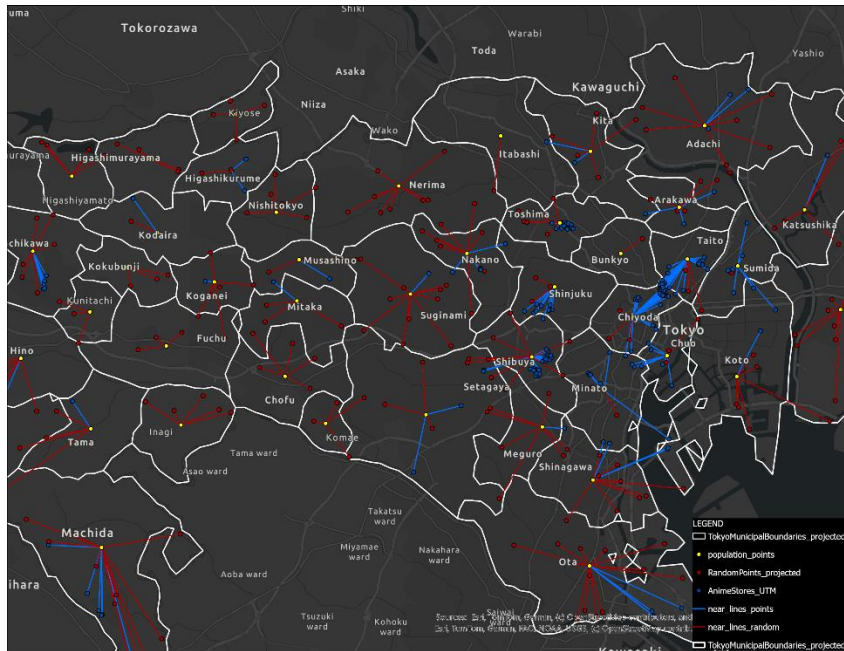
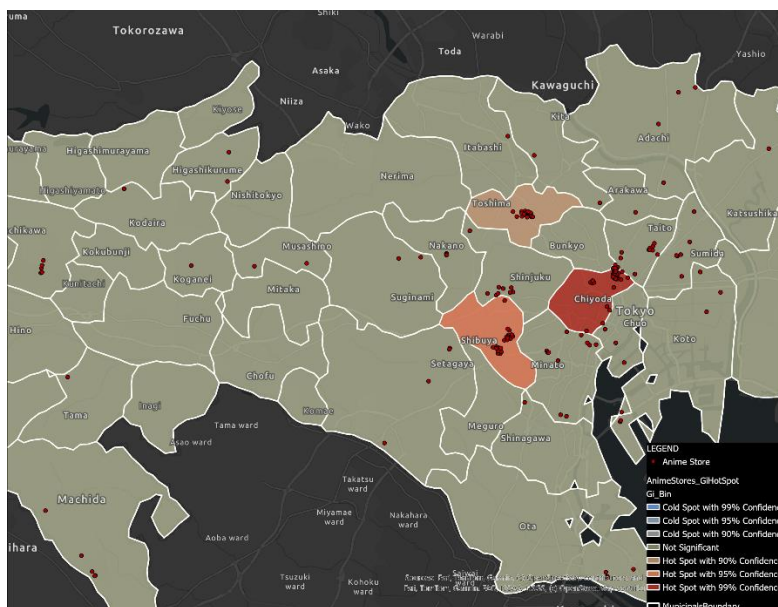


Figure 5. Getis-Ord Gi* hotspot analysis identifying statistically significant anime store clusters.



HERE IS THE LINK TO MY CODE:

https://docs.google.com/document/d/1rPmNx1lrjNZoV6rPq4pN6IQkBSssJbpHjWmFLY_NSdQ/edit?usp=sharing